

NEXUS OPERATIONS

Complete End-to-End User Guide & Operational Manual for
Administrators and Field Staff

ORGANIZATION

FES Fast Engineering Solutions

VERSION

Release 2.4 (Enterprise Edition)

SYSTEM NAME

NEXUS Operations Suite

TARGET AUDIENCE

Admins, Operations Managers, Field Staff

Purpose of this Document: This manual provides a clear, step-by-step walkthrough of how to use the NEXUS Operations web application. It covers everything from complaint email intake, surveys, quotations, and PO approvals to tax invoicing, work completion, field expense tracking, payments, and executive ledger reporting.

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1.0 System Overview & Core Philosophy

NEXUS Operations is an enterprise operational lifecycle management system built specifically for engineering solutions, facility management, and maintenance contractors. It replaces scattered spreadsheets, manual email threads, and paper records with a unified, transparent pipeline.

Key Benefits & Objectives

- **Zero Lost Complaints:** Ingests complaint emails directly from Gmail, assigns unique sequential serial numbers, and tracks resolution progress.
- **Standardized Documentation:** Generates uniform, branded PDF Quotations, Survey Reports, Invoices, and Completion Certificates in seconds.
- **Financial Precision:** Tracks costs, tax deductions at source (WHT), payments received, and exact job-level Profit or Loss in real time.
- **Simple Employee Access:** Employees log in with a fast 4–6 digit PIN without needing email setups.

2.0 User Access & Logging In

The system supports two distinct roles with tailored interfaces:

ROLE	LOGIN METHOD	PERMISSIONS & CAPABILITIES
Administrator LEVEL 5 ACCESS	Email + Password	• Admin Command Center & Financial Overview • Employee creation, PIN updates, and password resets • Two-way commenting/messaging with employees • Full access to all job stages, documents, and ledger exports • Gmail integration & auto-sync controls
Employee FIELD STAFF	Fast 4–6 Digit PIN	• Viewing Assigned Tickets and Intake • Submitting Survey Reports & Site Photos • Generating Quotations & Invoices • Recording Work Completion, Expenses & Payments Received • Replying to Admin Comments

STEP-BY-STEP LOGIN INSTRUCTIONS

1. Open the web app URL in any modern web browser (Google Chrome, Edge, Safari, or Firefox).
2. **For Admins:** Select the "Admin Login" tab, enter your registered email and password, and click **Sign In**.
3. **For Employees:** Select the "Employee PIN" tab, enter your employee name and your 4–6 digit PIN, and click **Log In**.

3.0 The 8-Stage Operational Lifecycle

Every maintenance job in NEXUS Operations follows a structured, sequential 8-stage journey. This ensures that no job moves to invoicing or completion without proper survey, quotation, and client approval.

1 Ticket Ingestion & Complaint Intake

Routes: </tickets> & </intake>

The intake process begins when customer complaint emails arrive or when a customer calls in an emergency.

How to Process Intake:

1. Navigate to **All Tickets Inbox** (</tickets>) in the sidebar.
2. Each ticket shows its sequential **Serial Number**, Date, Time, Sender, and Subject.
3. Click on any ticket to review its content. You can categorize it into one of four statuses:
 - **RELEVANT** : Valid job request that requires technical action.
 - **IRRELEVANT** : Spam, newsletter, or unrelated communication.
 - **CANCELLED** : Complaint cancelled by client.
 - **INTAKE DONE** : Already converted into an active job.
4. To turn a ticket into an active job, open **Intake** (</intake>):
 - **Client Name**: Commercial client name (e.g., Bank Alfalah, Allied Bank, Shell).
 - **Site / Branch**: Specific branch location (e.g., Mall Road Branch, F-7 Islamabad).
 - **Person of Contact (POC)**: Name and phone number of the site manager.
 - **Nature of Work**: Short summary of the issue (e.g., AC compressor replacement, Main breaker tripping).
 - **Assign Employee**: Select the field engineer responsible for handling this job.
5. Click **Save Job Intake**. The job is now registered in the system with an active tracking ID!

2 Survey & Field Assessment Report

Route: </survey>

After reaching the client's site, the assigned field engineer conducts an initial technical inspection.

How to Submit a Survey Report:

1. Navigate to **Survey Report** (</survey>).
2. Use the **Job Selector** dropdown at the top to select the client branch.
3. Type in the **Technical Findings & Observation Notes** (e.g., "Found 3-phase capacitor burnt due to voltage surge; requires replacement of 45uF capacitor and contactor switch").
4. Upload **Site Inspection Photos** as visual evidence.
5. Click **Submit Survey Report**. This document is automatically saved into the job's permanent record.

Based on the survey report, generate a formal commercial price quotation to send to the client for budget approval.

How to Create a Quotation:

1. Navigate to **Quotation Generator** (/quotation).
2. Select the target Job from the dropdown.
3. Add line items by specifying:
 - **Item Description:** Item name or service description (e.g., "Schneider 32A Contactor").
 - **Quantity:** Number of units or labor hours.
 - **Unit Price (Rs.):** Cost per unit.
4. Click **+ Add Line Item** for additional parts/labor.
5. The system automatically calculates the **Subtotal**, applies the configured **Tax Rate (e.g., 15%)**, and computes the **Net Quotation Total**.
6. Click **Generate & Save Quotation**. You can instantly preview, print, or download the formal PDF quotation to email to the client.

4 Purchase Order & Bank / Client Approval

Route: /approval

Before physical work or procurement starts, record the client's official authorization or PO confirmation.

How to Record Approvals:

1. Navigate to **Bank Approval & PO** (/approval).
2. Select the active Job.
3. Enter the **Purchase Order Number (PO #)** or Bank Reference Code.
4. Enter the **Approved Amount (Rs.)** authorized by the client.
5. Upload a photo or scan of the signed PO or bank approval email.
6. Click **Save Approval Record**. The job status updates to "Approved for Execution".

5 Commercial Tax Invoicing

Route: /invoice

When the job is authorized and ready for billing, create the formal commercial Sales Tax Invoice.

How to Generate an Invoice:

1. Navigate to **Invoice Studio** (/invoice).
2. Select the active Job. The system automatically pre-fills the client details and approved line items.
3. Review line items, invoice date, and payment terms (e.g., Net 30 Days).
4. Click **Issue Tax Invoice**.
5. The invoice is permanently archived with a unique Invoice Number and is instantly available for print/PDF export.

6 Site Work Execution & Completion

Routes: /site & /work-status

Tracks physical installation, maintenance, or repair progress at the client branch.

How to Update Work Status:

1. Navigate to **Site Work Status** (/site or /work-status).
2. Select the Job being executed.
3. Update execution status:
 - **PENDING** : Awaiting technician arrival / parts delivery.
 - **IN PROGRESS** : Work currently underway.
 - **WORK COMPLETED** : Physical work finished and tested on-site.
4. Upload **Work Completion Photos** and client site sign-off slip.
5. Click **Save Work Completion**.

7 Expenses & Client Payments Tracking

Route: </payment-status>

This stage handles real-time accounting for both money spent (field expenses) and money received from the client.

1. Recording Field Expenses (Costs):

- Select the active Job on the **Payment Status** page.
- Under **Add Expense**, enter the expense amount (Rs.), category/notes (e.g., "Purchased 15m 4mm Cable + Taxi fare"), and attach the receipt photo.
- Click **Save Expense**. The cost is instantly factored into the job's profitability calculation.

2. Recording Payments Received:

- Under **Payment Received**, enter the Amount Received into bank/cash.
- Enter the **Tax Deducted (WHT)** at source by the corporate client.
- Enter payment mode (Cheque #, Online Bank Transfer, Cash) and notes.
- Click **Record Payment**.
- The **Payment Progress Indicator** automatically updates from 0% to 100% (e.g., Partial Payment, Fully Paid).

8 Job Financial Ledger & Profit/Loss Audit

Route: </ledger>

The Job Ledger provides an executive 360-degree audit of all income and expenses for any job.

Ledger Breakdown Items

METRIC	DESCRIPTION
Total Quoted / Invoiced	Gross revenue billed to the client.
Total Tax Withheld (WHT)	Withholding tax deducted by corporate bank/client.
Net Receivables	Actual net cash expected after tax deduction.
Total Costs & Expenses	Sum of all field purchases, materials, and labor paid out.
Net Profit / Loss	(Total Received - Total Expenses) with visual Profit/Loss indicator.

4.0 Admin Command Center & Strategic Management

The **Admin Command Center** (`/admin/dashboard`) is the central nerve center for directors and operations managers. It provides top-level oversight across all company jobs, employees, and finances.

1. Strategic Workflow Matrix

Live, real-time counter tiles displaying where every complaint stands across the whole company:

- **Not Entered Complaints:** Raw emails not yet converted into jobs (Needs Attention).
- **Entered Emails:** Complaints successfully converted into active site jobs.
- **Generated Quotes:** Total formal price estimates prepared for clients.
- **Approved Quotes:** Quotes approved by bank/client with PO received.
- **Generated Invoices:** Billed jobs undergoing payment collection.

2. Live Employee Matrix & PIN Management

Admins can manage and monitor all team members from the **Employees** tab:

- **Add New Employee:** Click `+ Register User` , enter their name, and assign a 4–6 digit numeric PIN.
- **Change PIN:** Enter existing employee name with a new PIN to instantly update their login credentials.
- **Monitor Active Job:** See which ticket/serial number each employee is actively handling in the field.
- **Active/Inactive Toggle:** Safely grant or suspend access without deleting user history.

3. Dedicated Two-Way Employee Comments & Chat

Direct communication thread between administrators (Fatma/Anie) and individual field employees:

- Select any employee from the dropdown to open their private comment history.
- Type a message or instruction and click **Post Comment**.
- Employees see the admin note when they log in and can post replies directly.

5.0 All Documents Repository & CSV Exports

Accessible at `/admin/documents` , this master repository organizes all documents generated across the enterprise:

- **Document Filters:** Filter by Quotations, Invoices, Survey Reports, Bank Approvals, or Work Completions.

- **Date Range Filtering:** Specify *From Date* and *To Date* to audit performance for any fiscal month or quarter.
- **Download Ledger CSV:** Click **Download Ledger CSV** to export a clean spreadsheet formatted for external accounting software (QuickBooks, Excel, Zoho).

6.0 Gmail Integration & Email Sync Hub

Located at `/gmail`, this subsystem connects your company's operational inbox (e.g. `fesopreations@gmail.com`) to automatically ingest complaints.

HOW AUTO-SYNC WORKS

When complaint emails arrive in your Gmail, the system extracts the **Sender, Subject, Exact Date & Time**, assigns a **Sequential Serial Number**, and filters out spam or newsletters automatically.

Gmail Management Actions:

ACTION	WHEN TO USE
Sync Gmail Accounts Now	Regular button to fetch latest unread complaint emails from inbox.
Force Full Re-Sync	Re-scans all complaint emails from July 2026 onwards without deleting existing tickets or links.
Reconnect Account	Use when the Gmail account password has been changed. Re-authorizes Google OAuth in one click.
Run Diagnostics	Checks token expiry dates, Google API connectivity, and shows detailed sync health.

7.0 Frequently Asked Questions (FAQ) & Troubleshooting

Q1: I changed my Gmail account password and scanning stopped. What should I do?

Answer: When you change your Google password, Google automatically cancels previous app tokens. Go to `/gmail`, click **Reconnect** next to your account, sign in with your new password, and click **Allow**. Your scanning will resume immediately with all existing data preserved.

Q2: An employee forgot their PIN. How can they log in?

Answer: An admin can open the **Admin Command Center** (`/admin/dashboard`), go to the **Employees** tab, click **Register User**, enter the employee's name, and enter a new 4–6 digit PIN. The PIN is updated instantly.

Q3: An email was marked "Irrelevant" by mistake. How can we convert it to a Job?

Answer: Go to **All Tickets** (`/tickets`), click on the ticket, change its status pill to **Relevant**, then go to **Intake** (`/intake`) and enter the client branch details.

Q4: How do I export records for tax or accountant auditing?

Answer: Go to **All Documents** (</admin/documents>), select your date range (e.g., July 1 to Sept 30), and click **Download Ledger CSV**. The file can be opened directly in Microsoft Excel.